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28 February 1972
SUPERSEDING
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6 April 1970

MILITARY SPECIFICATION

BODY ASSEMBLY: 10541950

1. SCOPE

1.1 This specification covers a modular unit of the Tank Commander's M36 Periscope or the Tank Gunner's M32 or M35 Periscope, herein referred to as Body Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

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DRAWINGS

U.S. Army, Frankford Arsenal

F8270621	Test Fixture for Infrared Body Assembly, Periscopes: M32 and M36
F8298786	Support, Test Fixture Accessories
F8298789	Projector Collimating, Parabolic
F8565556	Pressure Tester, Two-Station
F8577787	Set-Up Instructions for Universal Test Stand
F10539179	Calibration Standard for Infrared Body Assembly, Periscopes: M32 and M36
A10539180	Set-Up Instructions for Test Fixture, F8270621, Infrared Body Assembly - M32 and M36
A10539181	Operating Instructions for Test Fixture F8270621, Infrared Body Assembly - M32 and M36
F10541950	Body Assembly: 10541950
F11729724	Gage Interchangeability, Maximum
F11732100	Gage Interchangeability, Minimum

PACKAGING DATA SHEET

10541950	Packaging of Body Assembly: 10541950

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The body assembly shall be manufactured in accordance with Drawing F10541950 and drawings pertaining thereto.

3.2 Interchangeability.- The body assembly shall be capable of insertion into and removal from the gages specified in 4.6.12.

3.3 General specifications.- The contractor shall be responsible for adherence to, and compliance with the following requirements of MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

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3.4 Environmental.

3.4.1 Storage temperature.- The body assembly shall show no evidence of cement separation, glass breakage or other physical failure, and shall meet all the requirements of this specification at standard ambient temperature (plus 60° to 90°F) after having been exposed and thermally stabilized at ambient temperatures of minus 80° to plus 160°F.

3.4.2 Operating temperatures.- The body assembly shall meet the applicable requirements of 3.5 while exposed and thermally stabilized at ambient temperatures of minus 40° and plus 150°F.

3.5 Torque.

3.5.1 Diopter movement.- The running torque required to rotate the diopter movement through the range specified in 3.6.4.7 shall be within the limits of 1.5 and 6.0 lb-ins when checked at standard temperature. When checked at the temperature conditions specified in 3.4.2, the running torque shall not exceed 15 lb-ins.

3.5.2 Focusing mechanism.- The running torque required to rotate Ring 8620059 through the range specified in 3.6.3 shall be within the limits of 4.0 and 14.0 lb-ins when checked at standard temperature. When checked at the temperature conditions specified in 3.4.2, the running torque shall not exceed 24.0 lb-ins.

3.5.3 Disengagement of boresight knob locks.- The direct force required to disengage each boresight knob from the lock position shall be within the limits of 2.0 to 5.0 pounds (lbs) when checked at standard temperature. When checked at the temperature conditions specified in 3.4.2, the direct force shall not exceed 9.0 lbs.

3.5.4 Rotation of boresight knobs.- The running torque required to rotate each boresight knob through the range specified in 3.6.4.3 shall be within the limits of 0.5 and 5.0 lb-ins when checked at standard temperature. When checked at the temperature conditions specified in 3.4.2 the running torque shall not exceed 8.0 lb-ins.

3.5.5 Electrical switch.- This requirement shall be met at standard temperature and the temperature conditions specified in 3.4.2. Knob MS91528-OE1B shall be manually operable to Position Switch 10516592 from the "OFF" to "24V ON" positions.

3.6 Performance.

NOTE: The following references apply throughout this specification. Datum Plane "X" and Reference Line "Y" as shown on Sheet 1 of Drawing F10541950.

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3.6.1 Dielectric strength (prior to installing image tube and power pack).- There shall be no puncture of insulation, no visible evidence of damage due to corona, or decrease in dielectric strength when a potential of 20,000 volts dc, plus or minus 500 volts, is applied between ground and image tube or power supply contact for a period of not less than one minute.

3.6.2 Electrical characteristics.

3.6.2.1 Image converter tube.- The image converter tube shall be energized by the application of +24 ± 6.0 vdc to the contact for the IR power supply (reference Drawing F10541950).

3.6.2.2 Reticle projector lamp.- The reticle projector lamp shall be energized by the application of +24 ± 6.0 vdc to the contact for the reticle lamp (reference Drawing F10541950).

3.6.2.3 Reticle illumination.- With a minimum of 18 and a maximum of 30 volts energizing the projector lamp, the reticle markings shall appear clearly defined when viewing through the eyepiece with the image tube energized (see 3.6.2.1).

3.6.2.4 Image steadiness.- There shall be no visible flicker, no movement of the image nor audible sparking when observed for 1 minute. This observation shall be made after the expiration of 15 seconds tube stabilization time.

3.6.3 Objective focusing range.- Rotation of Ring 8620059 shall cause the objective to bring targets from 200 meters to 1200 meters into focus.

3.6.4 Collimation.- These requirements shall be met prior to meeting the requirements of 3.6.5.

3.6.4.1 Previbration setting.- An incident line of sight shall be provided which is perpendicular to Datum Plane "X", within 0.2 mil. Coincidence of this line of sight with the instrument reticle shall be established by rotation of the boresight knobs.

3.6.4.2 Reticle movement.- Reticle (boresight cross) movement shall be measured against an external target from the collimated setting of 3.6.4.1 to meet the requirements of 3.6.4.2.1 and 3.6.4.2.2.

3.6.4.2.1 Elevation.- Clockwise rotation of the elevation knob shall move the reticle, in a direction towards the knob, a minimum of 7.0 mils from the collimated setting. Counterclockwise rotation of the elevation knob shall move the reticle, in a direction away from the knob, a minimum of 7.0 mils from the collimated setting.

3.6.4.2.2 Deflection.- Clockwise rotation of the deflection knob shall move the reticle, in a direction towards the knob, a minimum of 7.0

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mils from the collimated setting. Counterclockwise rotation of the deflection knob shall move the reticle, in a direction away from the knob, a minimum of 7.0 mils from the collimated setting.

3.6.4.3 Knob rotation.- Continuous movement of the reticle shall be evident until limited by positive engagement of clockwise and counterclockwise stops in each boresight knob.

3.6.4.4 Line of sight variation.- Variation in the line of sight in elevation, as the deflection knob is rotated by at least 7.0 mils in each direction from the collimated setting of 3.6.4.1, shall not exceed 0.1 mil (total variation not to exceed 0.2 mil).

3.6.4.5 Reticle shift.- Disengagement or reversal of either knob shall not cause the line of sight to shift by more than 0.1 mil in a direction perpendicular to the intended motion.

3.6.4.6 Boresight backlash.- Backlash in each boresight mechanism shall not exceed 0.10 mil.

3.6.4.7 Eyepiece focus.- The diopter scale index shall index zero within plus or minus 1/4 diopter when the screen is brought into best focus with the aid of a dioptometer. From this best focus the eyepiece shall have at least 3.0 diopter movement in both directions as read on the eyepiece scale.

3.6.4.8 Diopter backlash.- Backlash in the diopter movement shall not exceed 1/8 diopter.

3.6.4.9 Image tilt.- The image of a line originally parallel to Surface "X" and perpendicular to Reference Line "Y" shall emerge from the eyepiece parallel to the horizontal reticle line within 1.0 degree of arc.

3.6.4.10 Parallelism of reticle and image.- The image of a target line which is originally parallel to Surface "X" and perpendicular to Reference Line "Y" shall emerge from the eyepiece parallel to the horizontal reticle line within 1.0 degree of arc.

3.6.4.11 Resolution.- The resolution shall be 1 minute of arc or less. Resolution consists of detectable line structure and proper line count in all four meridians. Eyepiece focus shift between any two meridians shall not exceed 0.25 diopter.

3.6.5 Vibration.- This requirement shall be met with the image converter tube and reticle projector lamp energized in accordance with 3.6.2.1 and 3.6.2.2. The body assembly shall be vibrated in a vertical plane at a constant frequency of 30 cycles per second (cps) with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes plus or minus 15 seconds. There shall be no failure or flicker of light emission from the image converter tube and reticle projector lamp during

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vibration and the body assembly shall show no evidence of physical failure subsequent to vibration. Following the vibration test, the body assembly shall meet all remaining requirements herein.

3.6.6 Post vibration collimation.- The original collimation pre-vibration settings established in 3.6.4.1 shall not deviate more than 0.1 mil (including fixture error) due to vibration, as observed through the eyepiece.

3.6.7 Electron image shift.- After vibration, the IR body shall be subjected to ten cycles of "ON" and "OFF" rated voltage, the voltage being applied for 15 seconds and removed for 10 seconds. The collimation of the body assembly at the expiration of each "ON" period shall not deviate more than 0.10 mil.

3.6.8 Image centering.- An incident line of sight perpendicular to Datum Plane"X" (within 0.2 mil) shall appear centered in the eyepiece field of view within 5.0 degrees relative to the axis defined by the cylindrical diameter of the eyepiece diopter scale ring.

3.6.9 Cleanliness.

a. Defects.- There shall be no evidence of moisture, grease, fingerprints, condensates, fractures and adhesive separations on any glass component when viewing through either end.

b. Dirt.- When viewing through the eyepiece with the image converter tube energized, there shall be no more than five particles visible on the screen surface. The size of any particle shall not exceed the apparent width of the reticle line.

3.6.10 Sealing.- The body assembly shall show no evidence of leakage in excess of 0.2 pound per square inch gage (psig) when subjected to an internal pressure of 5 ± 0.10 psig for a minimum of 2 hours. The interior atmosphere of the body assembly shall contain nitrogen (having a dew point at least as low as minus 25°F at atmospheric pressure) between 0.25 and 0.50 pound per square inch gage (psig).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three body assemblies plus three each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. In addition to 100% inspection as specified in 4.3.2.2, the tabulated classification of defects in Tables I, II and III

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shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.2.2 100 Percent inspection.- The following requirements and related tests shall be subjected to 100% inspection:

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
1.	Image converter tube	3.6.2.1	4.6.4.2.1
2.	Reticle projector lamp	3.6.2.2	4.6.4.2.2
3.	Image steadiness	3.6.2.4	4.6.4.2.4
4.	Line of sight variation	3.6.4.4	4.6.4.7
5.	Reticle shift	3.6.4.5	4.6.4.8
6.	Eyepiece focus	3.6.4.7	4.6.4.10
7.	Resolution	3.6.4.11	4.6.5
8.	Interchangeability	3.2	4.6.12

NOTE: The tests in 4.3.2.2 shall be conducted at a temperature between + 60° and + 90°F.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL:	NONE		
MAJOR:	AQL 0.65% DEFECTIVE		
101.	Torque, diopter movement	3.5.1	4.6.3.1
102.	Torque, focusing mechanism	3.5.2	4.6.3.2
103.	Force, disengagement of boresight knob locks	3.5.3	4.6.3.3
104.	Torque, rotation of boresight knobs	3.5.4	4.6.3.4
105.	Torque, electrical switch	3.5.5	4.6.3.5
106.	Dielectric strength	3.6.1	4.6.4.1
107.	Reticle illumination	3.6.2.3	4.6.4.2.3
108.	Objective focusing range	3.6.3	4.6.4.3
109.	Collimation, previbration setting	3.6.4.1	4.6.4.4
110.	Reticle movement, elevation	3.6.4.2.1	4.6.4.5
111.	Reticle movement, deflection	3.6.4.2.2	4.6.4.5.1
112.	Knob rotation	3.6.4.3	4.6.4.6
113.	Boresight backlash	3.6.4.6	4.6.4.9
114.	Diopter backlash	3.6.4.8	4.6.4.11
115.	Image tilt	3.6.4.9	4.6.4.12
116.	Parallelism of reticle and image	3.6.4.10	4.6.4.13
117.	Vibration	3.6.5	4.6.6
118.	Post vibration collimation	3.6.6	4.6.7
119.	Electron image shift	3.6.7	4.6.8
120.	Image centering	3.6.8	4.6.9
121.	Cleanliness, dirt	3.6.9b	4.6.10

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TABLE I - CLASSIFICATION OF DEFECTS CONT'D

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
122.	Cleanliness, defects	3.6.9a	4.6.10.1
123.	Sealing	3.6.10	4.6.11

MINOR: NONE

NOTE: The tests in Table I shall be conducted at temperature between +60° and +90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One body assembly shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable Drawing - Visual
302.	General specifications	3.3	MIL-F-13926 - MIL-O-13830 Visual

NOTE: The tests in Table II shall be conducted at a temperature between 60° and 90°F.

4.4.2 Environmental.- Three body assemblies shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table III and shall then meet the requirements and tests in Table I.

TABLE III

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Storage temperatures (-80° to +160°F)	3.4.1	4.6.1
304.	Torque, diopter movement	3.4.2, 3.5.1	4.6.2, 4.6.3.1
305.	Torque, focusing mechanism	3.4.2, 3.5.2	4.6.2, 4.6.3.2
306.	Force, disengagement, boresight knob locks	3.4.2, 3.5.3	4.6.2, 4.6.3.3

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TABLE III CONT'D

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
307.	Torque, rotation of bore-sight knobs	3.4.2, 3.5.4	4.6.2, 4.6.3.4
308.	Torque, electrical switch	3.4.2, 3.5.5	4.6.2, 4.6.3.5

NOTE: The tests for Characteristics 304 through 308 inclusive, shall be conducted at -40° and at +150°F.

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted items have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- All inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the technical data package list.

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Special inspection equipment.

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4.5.2.3.1 Sealing.- Special testing equipment to test sealing of the body assembly shall conform to the internal pressure test requirements of Drawing F8565556, Pressure Tester-Two Station.

4.5.2.3.2 Performance.- Special test equipment to determine performance of the body assembly shall conform to the requirements of Drawing F8270621 (Test Fixture for Infrared Body Assembly, Periscopes M32 and M36), Drawing F8298786 (Support (Work Table), Test Fixture Accessories), Drawing F8298789 (Projector Collimating, Parabolic), Drawing F8577787 (Universal Test Stand and Set-Up Instructions) and Drawing F10539179 (Calibration Standard for Infrared Body Assembly, Periscopes M32 and M36).

4.6 Test methods and procedures.

4.6.1 Storage temperatures.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.1. The body assembly shall be placed in the test chamber and the temperature of the chamber reduced gradually (see NOTE below) to -80°F. The body assembly shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, the temperature of the test chamber shall be gradually raised (see NOTE below) to +160°F. The body assembly shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, the temperature of the test chamber shall be gradually reduced to room ambient temperature (+60° to +90°F), and the body assembly removed from the test chamber. The body assembly shall then be subjected to the visual, tactile, and audible examination of 4.6.6. The body assembly shall conform to the requirements of 3.4.1.

NOTE: The rate of temperature change in the test chamber shall not exceed 20 degrees per hour throughout the temperature cycling tests of 4.6.1 and 4.6.2.

4.6.2 Operating temperatures.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.2. The body assembly shall be placed in the test chamber and the temperature of the chamber reduced gradually (see NOTE in 4.6.1) to -40°F. The body assembly shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period and while at -40°F, the body assembly shall be subjected to the tests in 4.6.3.1 to 4.6.3.5 inclusive, to determine compliance with the applicable requirements of 3.5.1 to 3.5.5 inclusive, for this temperature (-40°F). Upon completion of this test, the temperature of the test chamber shall be gradually raised (see NOTE in 4.6.1) to +150°F. The body assembly shall remain at this temperature for a period of 5 hours. At the completion of this 5 hour period and while at +150°F, the body assembly shall again be subjected to the tests in 4.6.3.1 to 4.6.3.5 inclusive, to determine compliance with the applicable requirements of 3.5.1 to 3.5.5 inclusive, for this temperature (+150°F). Upon completion of this test, the temperature of the test chamber shall be gradually reduced (see NOTE in 4.6.1) to room

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ambient temperature (+60° to +90°F) and the body assembly removed from the test chamber.

4.6.3 Torque.

4.6.3.1 Diopter movement.- Standard torque measuring equipment capable of measuring the torque values specified in 3.5.1 shall be utilized for this test. An appropriate adapter, to accommodate the diopter knob shall be used with the torque measuring equipment. Torque shall be measured while rotating the diopter movement through the range specified in 3.6.4.7. The running torque required to rotate the diopter knob shall conform to the applicable requirements of 3.5.1 for the temperatures indicated.

4.6.3.2 Focusing mechanism.- Standard torque measuring equipment capable of measuring the torque values specified in 3.5.2 shall be utilized for this test. An appropriate adapter, to accommodate Ring 8620059 shall be used with the torque measuring equipment. Torque shall be measured while rotating the ring through the range specified in 3.6.3. The running torque required to rotate Ring 8620059 shall conform to the applicable requirements of 3.5.2 for the temperatures indicated.

4.6.3.3 Disengagement of boresight knob locks.- Standard measuring equipment capable of measuring the force values specified in 3.5.3 shall be utilized for this test. The force required to disengage each boresight knob from the lock position shall conform to the applicable requirements of 3.5.3 for the temperatures indicated.

4.6.3.4 Rotation of boresight knobs.- Standard torque measuring equipment capable of measuring the torque values specified in 3.5.4 shall be utilized for this test. An appropriate adapter, to accommodate the boresight knobs shall be used with the torque measuring equipment. Torque shall be measured while rotating each boresight knob through the range specified in 3.6.4.3. The running torque required to rotate each boresight knob shall conform to the applicable requirements of 3.5.4 for the temperatures indicated.

4.6.3.5 Electrical switch.- Manually operate the Knob 10516592. The knob shall meet the requirements specified in 3.5.5, at the temperatures indicated in 3.4.2 in addition to the standard ambient temperature (60°F to 90°F).

4.6.4 Performance.- The following tests shall be performed with the body assembly mounted on special testing equipment conforming to the requirements of 4.5.2.3.2. Set-up of the test equipment shall be in accordance with the set-up instructions contained on Drawings F8298789, F8577787 and A10539180. Mounting and securing of the body assembly on the test equipment shall be performed in accordance with the operating instructions contained on Drawing A10539181. Unless otherwise specified during these tests, the Collimator Projector 8298789 shall be adjusted for 1200 meters, the image converter tube and the reticle projector lamp shall be energized and the Infrared Focusing Ring 8620059 shall be adjusted for best focus of the target image after the diopter adjustment knob of the eyepiece is set for best focus of the phosphor grain on the image converter tube.

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The image converter tube shall be permitted a stabilization period of 15 seconds prior to conducting the optical tests of the infrared eight power system. A calibrated observation telescope of at least three power (3X) shall be a part of the test equipment. The observation telescope shall be positioned and mounted so that the optical axis of the telescope is in alinement with the line of sight emitting from the central aperture (geometric center) of the body assembly eyepiece. The reticle of the telescope shall have a crossline reticle pattern with tolerance lines to determine image tilt. The vertical reticle line of the telescope shall be adjusted plumb. The optical axis of the telescope shall be alined to the infrared target projector collimator and autocollimated to Datum Plane "X" within 10 seconds. Upon completion of the fixture set-up and alinement procedures, the observation telescope shall be capable of being removed from the optical line of sight of the test equipment when not required in the performance tests. Autocollimation and alinement shall be accomplished with the body assembly removed from the line of sight through the infrared target projector collimator and the observation telescope.

4.6.4.1 Dielectric strength.- The body assembly shall be tested in accordance with Method 301 of MIL-STD-202, utilizing standard measuring equipment. Dielectric strength shall be in conformance with requirements of 3.6.1. This is an in-process test that is performed with the image tube, transistorized power supply, voltage converter-regulator, and reticle projector lamp removed from the body assembly. The inspection records for the body assembly shall indicate the results of this test.

4.6.4.2 Electrical characteristics.

4.6.4.2.1 Image converter tube.- Apply the voltage specified in 3.6.2.1 to the contact for the IR power supply (reference Drawing F10541950) and position the electrical switch to the "24 volt" position. After allowing a 15 second tube stabilization period, view through the body assembly eyepiece and observe the image tube screen. The screen surface of the tube shall be illuminated.

4.6.4.2.2 Reticle projector lamp.- Perform this test concurrent with the test in 4.6.4.2.1. Apply the voltage specified in 3.6.2.2 to the contact for the reticle projector lamp (reference Drawing F10541950). The reticle pattern shall be projected into the optical system when observed through the body assembly eyepiece.

4.6.4.2.3 Reticle illumination.- Perform this test concurrent with the test in 4.6.4.2.2 and while the reticle projector lamp is energized at each voltage specified in 3.6.2.3. The reticle markings shall appear clearly defined at each required voltage setting when visually inspected through the body assembly eyepiece.

4.6.4.2.4 Image steadiness.- Perform this test with the reticle projector lamp off. Position the observation telescope at the eyepiece of the body assembly. View through the observation telescope and obtain coincidence of the horizontal and vertical reticle line and the image of the collimator reticle pattern lines on the screen of the image tube. Observe the target image for a time period of not less than one minute.

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During this one minute time period there shall be no visible flicker, no movement of the image nor audible sparking as specified in 3.6.2.4. Image movement shall be determined by means of the target image with respect to the observation telescope reticle pattern. Sparking shall be determined by means of an audio examination. Upon completion of this test, energize the reticle projector lamp in accordance with 3.6.2.2.

4.6.4.3 Objective focusing range.- The test for short range focus requirements of the body assembly shall be performed with the target image adjusted to a focus of 200 meters. With the diopter ring positioned for best focus of the phosphor grain on the screen of the image tube, rotate focusing ring of the body assembly to short range setting. The body shall conform to the requirements specified in 3.6.3. The test for long range focus requirements of the body assembly shall be performed with the target image adjusted for 1200 meters. Rotate the focusing ring to the long range setting to determine compliance with the requirements specified in 3.6.3.

4.6.4.4 Collimation, pre-vibration setting.- View through the body assembly eyepiece and obtain coincidence of the body assembly reticle boresight cross and the projected target image crosslines, by means of the boresight knobs. When coincidence has been obtained, slip the scales of the boresight knobs to a setting of 4 and recheck to insure coincidence is not disturbed. Record these settings for use in the test in 4.6.7. Without disturbing the boresight knob scale settings of 4, the body assembly shall then be subjected to the tests specified in 4.6.4.5 through 4.6.5 inclusive.

NOTE: Insure that knob scales are not disturbed (re-slipped) from their original setting of 4 throughout remainder of tests.

4.6.4.5 Reticle movement, elevation.- Establish coincidence of the body assembly reticle boresight cross and the projected target image crosslines by means of the boresight knobs (do not slip knob scales). While observing the boresight cross (intersecting point), rotate the elevation knob in the clockwise direction. The boresight cross shall move as specified in 3.6.4.2.1 as observed and measured on the target. Re-establish coincidence of the reticle boresight cross and the projected target image crosslines by means of the boresight knobs. While observing the boresight cross, rotate the elevation knob in the counterclockwise direction. The boresight cross shall move as specified in 3.6.4.2.1 as observed and measured on the target.

4.6.4.5.1 Reticle movement, deflection.- Perform this test utilizing the procedure in 4.6.4.5 except, the deflection knob shall be rotated in the clockwise and counterclockwise direction. Boresight cross (intersecting point) movements shall meet the requirements specified in 3.6.4.2.2 as observed and measured on the target.

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4.6.4.6 Knob rotation.- Perform this test in conjunction with the tests in 4.6.4.5 and 4.6.4.5.1. During rotation of each boresight knob (elevation and deflection), the boresight cross (intersecting point) shall be observed and show no evidence of irregularities (jump, creep or hang-up) in direction of travel until limited by engagement of each knob stop (clockwise and counterclockwise) to be in compliance with 3.6.4.3. While the boresight knobs are at each stop position, back off the knobs for at least 1/4 rotation and then again return the knobs to the stop. Perform this a minimum of 3 times. During these rotations, the knobs shall show no evidence of irregularity of movement (sponginess or friction) while approaching or reaching the stop. Contact with the stop shall be solid, such that a distinct metallic click can be heard to verify that reticle travel is being limited only by means of the knob stops and not by means of contact with adjacent components to determine overall compliance with 3.6.4.3.

4.6.4.7 Line of sight variation.- Perform this test in a manner similar to the test in 4.6.4.5.1. During rotation of the deflection knob, any variation in elevation in the line of sight, through the boresight cross (intersecting point), shall not exceed the values specified in 3.6.4.4.

4.6.4.8 Reticle shift.- With backlash removed for direction of travel, establish coincidence of the body assembly reticle boresight cross as in 4.6.4.5. While observing the intersecting point of the body assembly reticle boresight mark, disengage the elevation boresight knob from the lock position and proceed to rotate the knob (insure backlash is not introduced) for at least 1/4 turn, then reverse rotation of the knob (backlash included) for at least 1/4 turn. During the disengagement or reversal of the boresight knob, the line of sight through the body assembly reticle boresight mark shall not shift (jump) in a direction perpendicular to the intended motion more than the tolerance specified in 3.6.4.5. In addition to checking shift at this setting, repeat the above at the midpoint and maximum point of travel in each direction for a total of 5 checks overall. These 5 checks shall also be conducted with the deflection knob in compliance with 3.6.4.5.

4.6.4.9 Boresight backlash.- While viewing through the body assembly eyepiece, disengage and rotate the boresight knobs (insure backlash is removed for direction of travel) until a line of sight, through the intersection of the body assembly reticle boresight mark is positioned coincident with the intersecting point of the target image crosslines. With the boresight knobs in the lock position, record the boresight knob scale readings and the position of the line of sight. Disengage the boresight knobs and continue rotation approximately 1/2 of a revolution then return and engage the knobs to their original lock position as recorded above. View through the eyepiece and record the position of the line of sight through the reticle boresight mark. Any difference between the recorded positions of the line of sight for each boresight knob is the amount of backlash. Backlash in each boresight mechanism shall not exceed the tolerance

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specified in 3.6.4.6, as measured on the target.

NOTE: Should overtravel occur during rotation of the boresight knobs, the test is nullified and shall be repeated.

4.6.4.10 Eyepiece focus.- A calibrated dioptometer with a magnification of at least 3 power (3X) shall be utilized for measuring the accuracy of the zero diopter setting of the diopter scale. Set the dioptometer objective scale to zero and adjust the dioptometer eyepiece for best focus of the dioptometer reticle. Set the diopter knob of the body assembly eyepiece to indicate zero diopter. Position the dioptometer at the eyepiece of the body assembly. While viewing through the dioptometer, obtain best focus of the phosphor grain on the image tube by adjustment of the body assembly diopter knob. The phosphor grain of the image tube shall be in best focus within the tolerance specified in 3.6.4.7 as measured on the diopter scale of the body assembly. Rotate the diopter adjustment knob from the established best focus of the target image to its clockwise stop, note the diopter setting on the scale, then rotate the diopter knob to its counterclockwise stop, note diopter setting on the scale. The diopter movement shall meet the requirements specified in 3.6.4.7, as measured on the diopter scale of the body assembly.

4.6.4.11 Diopter backlash.- With backlash excluded for direction of travel, rotate the diopter knob clockwise and obtain best focus of the phosphor grain on the screen of the image tube. Record the diopter scale reading. Continue rotation of diopter knob to its stop, then with backlash included, rotate the knob counterclockwise and obtain best focus of the phosphor grain on the screen of the image tube without overtravel. Record the diopter scale reading. The difference between the first and second recorded scale reading is the amount of backlash in the diopter movement. Backlash in the diopter movement shall not exceed the tolerance specified in 3.6.4.8.

NOTE: Should overtravel occur during rotation of the diopter knob the test is nullified and shall be repeated.

4.6.4.12 Image tilt.- Position an auxiliary telescope containing a crossline reticle pattern at the body assembly eyepiece. The vertical reticle line of the auxiliary telescope shall be aligned plumb when in the test position on a qualified surface. With aid of the auxiliary telescope, view through the eyepiece of the body assembly and establish one extremity of the telescope vertical reticle line coincident to the vertical target image line. The image of the vertical target line shall be parallel to the telescope vertical reticle line within the tolerance specified in 3.6.4.9 as measured on the target.

4.6.4.13 Parallelism of image and reticle.- Perform this test while observing through the body assembly eyepiece with the unaided eye. Position one extremity of the body assembly vertical reticle line coincident to the vertical target image line. The body assembly vertical reticle line shall be parallel to the vertical target image line within the tolerance specified in 3.6.4.10, as measured on the target.

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4.6.5 Resolution.- Remove the body assembly from the testing equipment. The test for resolution shall be performed utilizing a supporting medium capable of receiving the body assembly and a resolving power chart or a wall target representing the angular subtense for minutes of arc specified in 3.6.4.11. Construction of the resolving power chart shall be as specified in the "Resolution Test" of MIL-O-13830 and contain four lines sets as shown on Figure 1 therein. Disregard center numeral. The target shall be illuminated by means of a filtered tungsten lamp. The filter shall not pass visible light rays below 0.65 microns. Place the body assembly on the supporting medium and position it in a horizontal plane. Energize the image tube of the body assembly from a power source capable of supplying and maintaining 24 volts DC. The image tube shall be permitted a stabilization period of 15 seconds prior to conducting the resolution test of the infrared optical system. Upon completion of the stabilization period, view through the body assembly eyepiece with the unaided eye and obtain best focus of the resolving power chart by means of the eyepiece diopter adjustment knob. Any change in the diopter readings by adjustment of the eyepiece diopter knob to obtain the detectable line structure and proper line count in all four meridians shall be recorded, as measured on the knob scale. The difference between the recorded readings for any two meridians shall not exceed the value specified in 3.6.4.11. Upon successful completion of this test, the body assembly shall be subjected to the vibration test in 4.6.6.

4.6.6 Vibration.- When the body assembly is provided as part of a complete periscope, this test and the related test of 4.6.7 will not be required. This test shall be performed on body assemblies procured as a repair part item. Standard vibrating equipment capable of providing the amplitude and frequency specified in 3.6.5 shall be utilized for this test. Unless otherwise specified, the test equipment shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The body assembly shall be clamped on the vibrating equipment in its normal mounting or operating position (as depicted in the front view on Sheet 1 of Drawing F10541950) by means of an appropriate adapter. The vibration motion shall be in the vertical plane. Prior to vibration, energize the image converter tube and reticle projector lamp in accordance with 3.6.2.1 and 3.6.2.2. After allowing a 15 second tube stabilization period, the screen surface of the tube shall be illuminated and the reticle pattern shall be projected into the optical system when observed through the body assembly eyepiece. The body assembly shall then be vibrated for the time duration specified in 3.6.5. During the first, third and fifth minute of the vibration cycle, observe the screen surface of the tube and the reticle pattern. During these observations there shall be no failure or flicker of light emission from the image converter tube and reticle projector lamp. Immediately following the vibration cycle, and prior to the examination for physical defects resulting from poor workmanship and materials, the body assembly shall be subjected to the tests in 4.6.7 to 4.6.10.1 inclusive. Do not disturb the boresight knob scale settings of 4 during transfer of the body assembly. Subsequent to the test in 4.6.10.1 the body assembly shall be thoroughly examined

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for physical defects resulting from poor workmanship and materials to determine compliance with 3.6.5, prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts. With the exception of the slip scales, a tactile examination shall be made for functioning of moving parts (boresight knobs, boresight knob scales, etc.) for undue irregularities of movement, friction or looseness. An audible examination shall also be made of the body assembly for loose internal optical and mechanical components. Any one of the possible defects defined above shall be cause to consider a body assembly defective. Body assemblies which have passed the above examinations shall be subjected to the remaining tests specified in 4.6.11 through 4.6.12 inclusive.

4.6.7 Post vibration collimation.- Subsequent to the vibration test, the body assembly shall be positioned as in 4.6.4. Observe through the body assembly eyepiece and adjust by means of the boresight knobs to obtain coincidence of the body assembly reticle boresight cross with the target image. Boresight knob adjustments shall not exceed the tolerance specified in 3.6.6 as measured on the boresight knob scales.

NOTE: Do not slip knob scales.

4.6.8 Electron image shift.- Observe through the body assembly eyepiece and establish coincidence of the body assembly reticle boresight cross and the projected target image crosslines by means of the boresight knobs. When coincidence has been established, proceed with the test as described in 3.6.7. During each "ON" cycle, the projected target image with respect to the body assembly reticle shall be observed for repeatability of the coincident setting established above. At the expiration of each "ON" cycle, any adjustment of the boresight knobs to re-establish coincidence of the reticle boresight cross and the projected image crosslines shall not exceed the value specified in 3.6.7, as measured on the boresight knob scales.

4.6.9 Image centering.- Perform this test with the reticle projector lamp off. Affix an auxiliary telescope of at least three power to the eyepiece of the body assembly under test. The reticle of the telescope shall have a crossline reticle pattern with a circle which radially represents the 5.0 degrees tolerance. View through the auxiliary telescope and observe the projected target line image. The line of sight through the intersecting point of the vertical and horizontal target line image must be within, or in coincidence with, the circle on the auxiliary telescope reticle to appear centered in the eyepiece field of view within the value specified in 3.6.8. Upon completion of this test, energize the reticle projector lamp.

4.6.10 Cleanliness, dirt.- The reticle projector lamp and the image converter tube shall be energized during this test. Prior to performing the test, the image tube shall be permitted a stabilization period of 15 seconds. Subsequent to the tube stabilization period, obtain best

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focus of the phosphor grain on the screen of the image tube by adjustment of the eyepiece diopter knob. The screen surface of the image tube shall then be visually inspected for cleanliness in accordance with 3.6.9b and shall conform to the requirements specified therein. "Cold" and "Hot" spots on the screen surface of the image tube shall not be considered dirt.

4.6.10.1 Cleanliness, defects.- Remove the body assembly from the testing equipment. The reticle projector lamp and the image converter tube shall be de-energized for this test. With a two 15 watt desk fluorescent lamp held close to the objective window and the eyelens, view by reflected light the glass components of the body assembly utilizing the technique of shadowing to determine compliance with the cleanliness requirements of 3.6.9a.

CAUTION: Prolonged exposure of the image tube to direct rays of white light shall be avoided.

4.6.11 Sealing.- The Cap Assembly 8589920-1 and Cap 10515991 shall be removed from the body assembly during the sealing test. This portion of the test shall be performed utilizing a pressure gage capable of reading between 0.25 to 0.50 pound per square inch gage (psig) and a dew point tester. The following test sequence shall be adhered to. Check the internal pressure of the body assembly utilizing the pressure gage specified herein. Then, test the dew point of the interior atmosphere of the body assembly by means of the dew point tester. Interior atmosphere of the body assembly shall conform to the requirements of 3.6.10. Upon satisfactory completion of this portion of the test, internal pressurizing of the body assembly shall be performed with special testing equipment conforming to 4.5.2.3.1. Pressurize the body assembly with dry nitrogen gas, having a dew point at least as low as minus 25 degrees Fahrenheit, to 5 pound per square inch (psi), in accordance with the procedure for internal testing on Drawing F8565556. Retain the applied pressure in the body assembly for the time duration specified in 3.6.10. During this time period, the boresight knobs and the diopter adjustment shall be rotated to their clockwise and counterclockwise stops. At the end of this time period there shall be no evidence of leakage as indicated on the pressure gage of the testing equipment, to determine compliance with 3.6.10. Upon satisfactory completion of the sealing test, the body assembly shall then be flushed with the dry nitrogen gas specified herein. The interior atmosphere of the body assembly shall be periodically sampled through the dew point tester until the atmosphere shows a dew point at least as low as minus 25 degrees Fahrenheit. When this atmosphere is attained within the body assembly, the body assembly shall conform to the requirement of 3.6.10.

4.6.12 Interchangeability.- This test shall be performed utilizing the maximum and minimum interchangeability gages in Drawings F11729724 and F11732100 respectively, to determine compliance with 3.2.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10541950. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Body Assembly: 10541950, is a modular unit of the Tank Commander's Periscope: M36 or the Tank Gunner's Periscope: M32 or M35, primarily used in the fire control system of the 105MM, Gun Tank: M60 Series.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Applicable packaging data sheet.
(c) Selection of applicable levels of preservation, packaging and packing.
(d) Applicable stock number.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1240-A346

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of this specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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